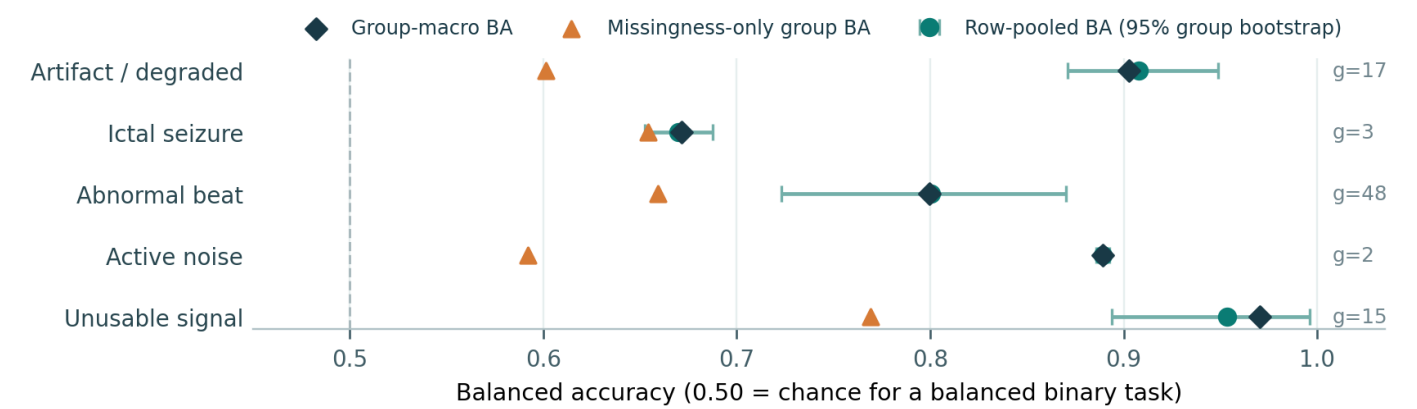


Can the feature matrix distinguish the clinical labels?

A four-page interpretation of 39,664 ECG feature rows using patient/source-grouped out-of-fold testing

Bottom line. The system did *not* predict everything correctly. Across the five selected tasks it made 7,162 target-level errors. Artifact and unusable-signal discrimination were strong internally; abnormal-beat discrimination was moderate; seizure discrimination was weak; and the noise estimate is based on only two independent source patients. None is an external clinical validation.



Selected out-of-group result for each target. The five target denominators overlap; 46,038 target-level predictions arise from 39,664 unique feature rows.

Target	Selected model and feature branch set	<i>g</i>	Sens.	Spec.	PPV	Group BA	Errors
Artifact / degraded	Logistic regression; B2 morphology + B4 quality	17	87.9%	93.7%	94.5%	90.3%	1,077
Ictal seizure	Linear SVM; B1 rhythm/HRV	3	63.0%	71.0%	19.7%	67.1%	3,388
Abnormal beat	Histogram boosting; B1+B2+B3	48	69.9%	90.2%	64.3%	80.0%	2,171
Active noise	Linear SVM; B2+B4	2	82.2%	95.7%	97.9%	88.9%	395
Strict unusable	Linear SVM; B2 morphology	15	92.7%	98.0%	88.0%	97.0%	131

How to read the statistics

Sensitivity: of all true positive episodes/rows, the fraction detected: $TP/(TP + FN)$. A low value means clinically relevant misses.

Specificity: of all true negatives, the fraction correctly rejected: $TN/(TN + FP)$. A low value means excess alarms.

PPV (precision): of all positive alerts, the fraction truly positive: $TP/(TP + FP)$. It depends strongly on prevalence.

Balanced accuracy (BA): $\frac{1}{2}(\text{sensitivity} + \text{specificity})$. It gives equal weight to both classes and is more informative than ordinary accuracy in an imbalanced dataset.

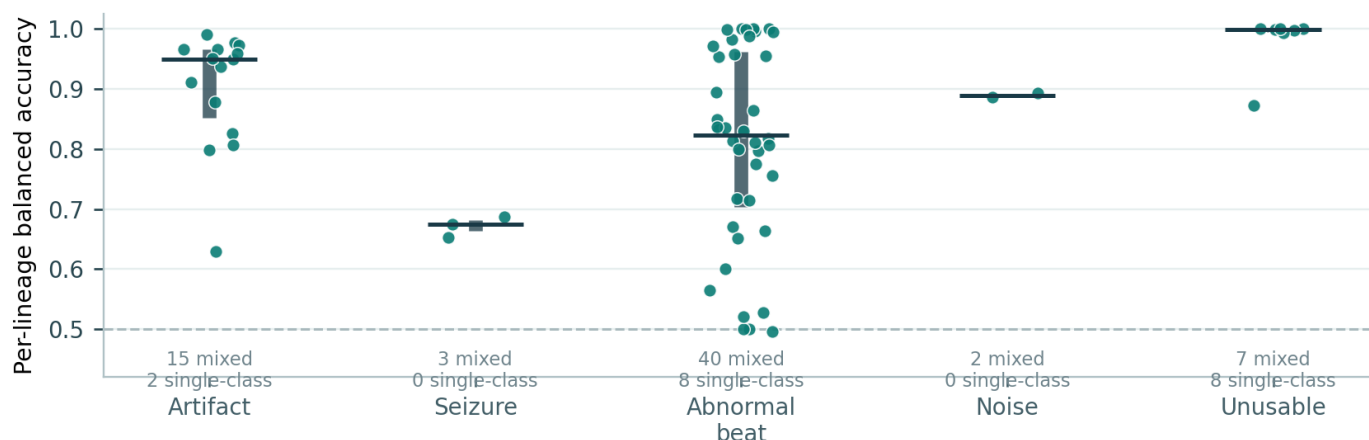
Group-macro BA: sensitivity and specificity are first calculated per independent lineage, then averaged. A long recording cannot dominate many shorter patients.

95% group bootstrap: 1,000 resamples of whole lineages, not correlated beats. The plotted interval is for row-pooled BA; the diamond is group-macro BA.

Audit design: 5 targets $\times$ 5 classifier families $\times$ all 15 non-empty branch combinations = 375 experiments. Every displayed prediction came from a fold that excluded its complete lineage group; preprocessing was fitted inside the training fold.

Group clusters: where performance is stable or fragile

Each point is one independent patient/source lineage with both positive and negative examples; bars show the interquartile range and median



Distribution across independent lineages. “Single class” means that BA is mathematically undefined because the group contains only positives or only negatives.

Target	All <i>g</i>	Mixed <i>g</i>	Median BA	IQR	BA < .60	BA ≥ .80
Artifact	17	15	95.0%	85.2–96.6%	0	13
Seizure	3	3	67.4%	66.3–68.1%	0	0
Abnormal beat	48	40	82.3%	70.3–96.1%	6	24
Active noise	2	2	88.9%	88.7–89.1%	0	2
Strict unusable	15	7	99.9%	99.5–100%	0	7

Three evidence clusters

- Broader but heterogeneous internal evidence - abnormal beats.** Forty-eight independent lineages are represented, yet six of the 40 mixed-class lineages score below 60% BA. The aggregate 80.0% group BA therefore hides clinically important record-to-record variation.
- Limited internal evidence - artifact and unusable signal.** Artifact covers 17 groups and is usually stable, but the worst mixed group is 63.0% BA. Unusable signal is excellent in seven mixed groups, while eight other groups contain only one class and cannot test both sensitivity and specificity.
- Critical sample-size limitation - seizure and active noise.** Seizure has only three patients/source groups and no group reaches 80% BA. Active noise appears strong but comes entirely from MIT-BIH source patients 118 and 119. Two similar patients cannot establish generalization.

Why the seizure number is not reassuring. The seizure classifier finds 728 of 1,156 ictal rows (63.0% sensitivity) but produces 2,960 false-positive rows. PPV is only 19.7%: about four of five positive alerts are false in this selected corpus. A feature-missingness-only model reaches 65.4% group BA, close to the selected model’s 67.1%.

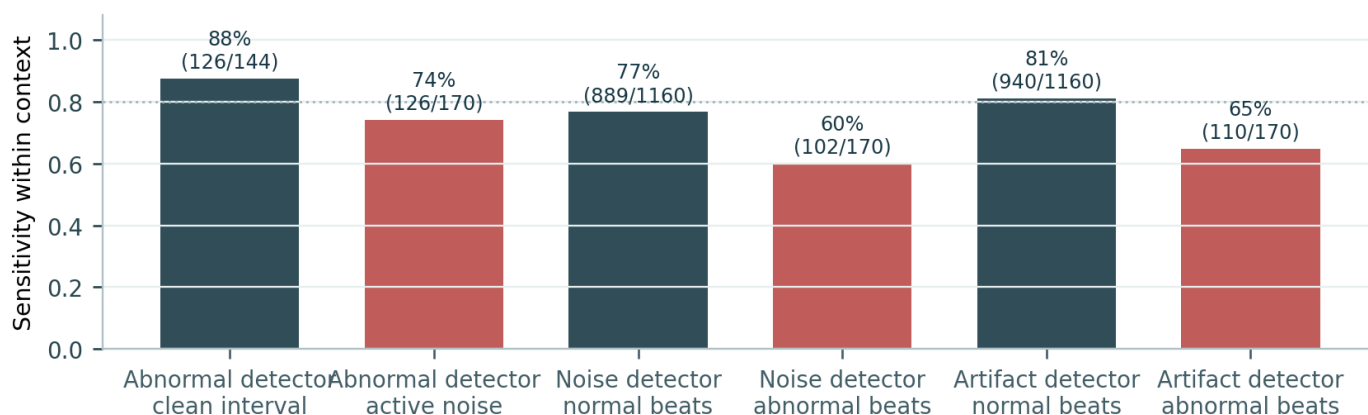
What the four feature branches mean

B1 rhythm/physiology (RR, heart-rate variability, PRSA/BPRSA); **B2 morphology** (template similarity and P-QRS-T shape); **B3 conduction/repolarization** (PR, QT, JT and related intervals); **B4 quality** (flatness, spectral ratios, detector agreement and template support). The audit tested every non-empty combination, but the group clusters above - not the winning branch label alone - determine how much confidence is warranted.

Negative controls: mean BA after 20 training-label permutations was 48.5% abnormal, 48.2% artifact, 54.2% noise, 50.3% seizure and 49.1% unusable. Near-chance permutations are necessary evidence against direct leakage, but they do not prove external validity.

Mixed clinical contexts: what can and cannot be measured

Joint labels exist for noise, artifact and beat type in NSTDB; seizure-plus-noise/artifact/beat combinations are not annotated



Clinically important interaction. In NSTDB, abnormal-beat sensitivity falls from 88% in clean intervals (126/144) to 74% during active noise (126/170). Among truly noisy rows, noise sensitivity falls from 77% on normal beats (889/1,160) to 60% on abnormal beats (102/170). Artifact sensitivity similarly falls from 81% to 65%. Abnormal morphology and noise can therefore mask one another.

Observed joint classes in NSTDB

Context	Normal beat	Abnormal beat	Total
Clean interval	684	144	828
Active noise/artifact	1,160	170	1,330
Total	1,844	314	2,158

These 2,158 rows have both an expert-matched beat class and the official noise schedule. The remaining 689 NSTDB rows retain noise/artifact truth but have no attached abnormal-beat target. Unknown labels were excluded, not changed to class 0.

The artifact and noise labels are identical in NSTDB by design: official inactive interval = 0 and active interval = 1. They are *different models*, however, because the artifact target is also trained on BUT-QDB quality labels.

Quality-severity classes in BUT-QDB

The composite artifact analysis contains 4,221 class-1 usable rows, 3,593 class-2 degraded rows and 656 class-3 unusable rows. The artifact model correctly rejects 96.1% of class 1, detects 87.8% of class 2 and detects 94.2% of class 3. The stricter unusable task compares class 1 versus class 3 only; class 2 is deliberately excluded because “degraded” is not the same clinical endpoint as “unusable.” The official three-class definitions and 15-subject structure are documented by [PhysioNet BUT-QDB](#).

Why seizure plus noise/abnormal/artifact is marked “not evaluable”

Requested combination	What the present data contain	Defensible conclusion
Seizure + active noise	Seizure intervals in CHB-MIT/Siena; no synchronized NSTDB noise truth	Not evaluable
Seizure + abnormal beat	Seizure intervals; no expert AAMI beat labels in the seizure rows	Not evaluable
Seizure + artifact/unusable	Seizure intervals; no BUT-QDB quality consensus in those recordings	Not evaluable
Noise + abnormal beat	Both labels coexist in 2,158 NSTDB rows	Evaluated above
Artifact + unusable	Both labels coexist in pure BUT-QDB class 1/3 windows	Severity analysis

Do not read “not evaluable” as zero. It means the necessary joint ground truth was never recorded in the same observations. Training a model to invent those missing labels would create an apparently complete but scientifically invalid matrix.

Case studies: concrete successes, misses and false alarms

Record-level results make the aggregate percentages clinically interpretable

Task	Record / group	Exact result	What failed	Clinical reading
Seizure	PN06_5 / PN06	1,079 rows: TP 22, FN 49, TN 681, FP 327; BA 49.3%; PPV 6.3%	Most ictal rows were missed and many non-ictal rows alerted.	A near-chance record despite the aggregate 67.1% group BA; unsuitable for alarm use.
Artifact	NSTDB 118e24 / 118	233 rows: TP 41, FN 116, TN 72, FP 4; BA 60.4%	Sensitivity only 26.1%; the model was conservative.	A clean-looking negative output cannot reliably exclude artifact in this record.
Active noise	NSTDB 118e24 / 118	233 rows: TP 66, FN 91, TN 65, FP 11; BA 63.8%	Sensitivity 42.0% despite 85.5% specificity.	The overall 88.9% hides a clinically weak record/window selection.
Abnormal beat	MIT-BIH 207 / 207	103 abnormal rows: TP 36, FN 67; sensitivity 35.0%	Two thirds of abnormal anchors were classified normal. Specificity is undefined because this selected window has no negatives.	Demonstrates why single-record accuracy and pooled specificity must not be substituted for sensitivity.
Strict unusable	BUT 111001 / 111	289 rows: TP 137, FN 47, TN 105, FP 0; BA 87.2%	No false unusable alarms, but 25.5% of unusable rows were missed.	Strong aggregate behavior still permits a clinically meaningful false-reassurance mode.

Strengths demonstrated by the audit

- The raw inventory byte-read and SHA-256-hashed 4,856 files (8.15 GB) with zero read errors. Six isolated reconstructions reproduced all 39,664 row IDs, all 52 feature values and all 56 semantic label fields with zero differences.
- Ten separate label-semantic checks found zero mismatches. Exact duplicate feature vectors never crossed lineage groups. All 375 experiment cells retained complete out-of-fold predictions.
- Artifact was strong in most mixed groups (median BA 95.0%; 13/15 groups at least 80%). Abnormal-beat performance was useful in many lineages (24/40 at least 80%) but clearly heterogeneous.

Defects and boundaries that remain

- **No external cohort.** The displayed winner was selected on the same cross-validation matrix. It is an inspection winner, not an unbiased final-test estimate.
- **Curated rather than full-waveform evidence.** The matrix contains 336 label-rich segments. A separate plan covers 557.81 raw signal-hours and 11,337 windows, but unlabelled windows were not fabricated as negatives.
- **Beat-classification boundary.** Abnormal truth is attached only when a pipeline beat anchor lies within 75 ms of an expert beat. Expert beats missed by the detector produce no classifier row, so this is not end-to-end detection sensitivity.
- **Shortcuts remain possible.** Missingness-only group BA is 65.4% for seizure and 76.9% for unusable signal. Source acquisition and feature availability can carry label information.

Clinical interpretation. These results justify continued engineering evaluation of artifact, morphology and abnormal-beat features. They do not justify diagnosis, seizure alarms or automated signal rejection. The next valid test requires patient-unique external data with joint seizure, beat-type and signal-quality annotation, plus an end-to-end accounting of missed detector beats.